Richard Lin

U.S. Citizen

Huntingdon Valley, PA | 215-403-4999 | richard.rich.lin@gmail.com
github.com/richardlin05 | richardlin05.github.io | linkedin.com/in/richardlin05

EDUCATION

Temple University Philadelphia, PA
Bachelor of Science in Electrical and Computer Engineering *Diploma Expected: May 2028*
Minor: Computer Science | Fundamentals of Physics Certificate | GPA: 3.88/4.00 | Dean's List: F24, S25, F25
Relevant Courses: Classical Control Systems; Signals and Systems; Processor Systems; Engineering Computation I-II-IV

PROFESSIONAL EXPERIENCE

Temple University Philadelphia, PA
Undergraduate Research Assistant – Computer Fusion Lab (CFL) *Oct 2025 – Present*
Advisor: Dr. Li Bai

- Administered lab website and systems; ensured security, software maintenance and network reliability
- Built a RAG + LLM-rerank recommendation agent within a multi-agent LLM system in Python, with a cron-automated pipeline syncing a 400+ device catalog daily and matching natural-language queries for people with disabilities
- Trained a Vision-Language-Action (VLA) model with LeRobot to enable a robotic arm to pick up and hand objects for individuals with limited mobility; separately implemented ROS2-based robot control in C++ across multiple robots

Undergraduate Teaching Assistant *Aug 2025 – Present*

- Assisted faculty with instruction, hands-on activities, and assignment evaluation; held office hours for student support
- **Intro to Engineering:** Guided students building embedded Micro:bit incubator, testing components (LEDs, fans, sensors)
- **Engineering Problem Solving:** Supported wind turbine calculation/data analysis using Excel, Python (NumPy, Matplotlib)
- **Engineering Computation I:** Taught C/C++/Python/Linux/Bash fundamentals, plus basic algorithms and time complexity

STEPS Ambassador – Sustainable Temple Energy and Power Scholars (STEPS) *Sep 2024 – Present*

- Mentored STEPS underclassmen through the high-school-to-college transition, providing academic guidance
- Designed and led renewable-energy STEM demos for high schoolers, emphasizing sustainability and real-world engineering

PROJECTS

Musical Light Show – ECE 2352: Circuits and Electronics II *May 2026*

- Designed analog audio-processing system that visualizes music as an audio-reactive LED light show
- Built op-amp mixer, potentiometer volume control, high/low-pass filters, peak detectors, BJT LED driver circuits
- Calculated values with cutoff-frequency/current-limiting equations; verified against Multisim and hardware, within $\sim$1%

Detective Conan AI Glasses (Childhood Dream Project) *Jun 2026 – Present*

- Architecting AI smart glasses on Raspberry Pi Zero 2 W and ESP32 with camera, mic, solar charging, 3D-printed enclosure
- Designing software architecture: intent engine with local/cloud LLM routing, object detection, OCR, cybersecurity tools

CLUBS & LEADERSHIP

Institute of Electrical and Electronics Engineers (IEEE) – Temple University Philadelphia, PA
President *Sep 2024 – Present*

- Led the $\sim$11-member IEEE executive board in planning outreach programs, technical workshops, and industry networking events, while coordinating with four affiliated chapters and societies (EMBS, PES, WIE, IEEE-HKN)
- Grew IEEE SAC participation by 50% (14 → 21) through targeted event promotion and outreach during 2025–2026
- Founded the first annual multi-university hardware hackathon with Drexel and UPenn IEEE branches

Eta Kappa Nu (IEEE-HKN) – Temple University Philadelphia, PA
Secretary & Treasurer *Jun 2026 – Present*

- Manage IEEE-HKN Headquarters correspondence, chapter finances, and induction/member events

Temple University College of Engineering Alumni Association Philadelphia, PA
Student Representative *Jun 2026 – Present*

- Served as liaison between the Alumni Association and students, connecting alumni for mentorship and networking

SKILLS

Programming & Web: Python, C, C++, Java, MATLAB, Bash, Verilog/SystemVerilog, JavaScript, HTML/CSS, UI/UX Design
AI & Robotics: LLM Integration, RAG, VLA, Edge AI, Computer Vision (OpenCV, YOLO), TTS/STT, Multi-Agent Systems, ROS
Hardware & Tools: Git/GitHub, Linux, Raspberry Pi, Arduino, FPGA, Circuit Design & Analysis, Soldering, n8n, PCB Design
Languages: English, Mandarin Chinese, Fuzhounese, Spanish